



Conditional Statements



If syntax

```
if condition:  
    code to execute when condition is true
```

If else Syntax

```
if condition:  
    code to execute when condition is true  
else:  
    code to execute when condition is false
```



Ladder if

```
if condition1:  
    code to execute when condition1 is true  
elif condition2:  
    code to execute when condition2 is true  
elif condition3:  
    code to execute when condition3 is true  
else:  
    code to execute when none of the conditions are true
```



Three Shapes of a Decision

Same idea — check a condition — three different structures depending on how many outcomes you need.

SIMPLE if

Runs a block only when the condition is True. Nothing happens otherwise.

```
if condition:
    do_something()
```

One path, one outcome.

if - else

Two possible paths. One branch always runs — True or False.

```
if condition:
    do_A()
else:
    do_B()
```

Exactly two outcomes.

if - elif ladder

Checks conditions top to bottom, stops at the first True match.

```
if c1:
    do_A()
elif c2:
    do_B()
else:
    do_C()
```

Many possible outcomes.



SIMPLE if

Q1. Write a program that checks whether a number entered by the user is positive.

```
num = int(input("Enter a number: "))  
  
if num > 0:  
    print("The number is positive")
```

SAMPLE OUTPUT

```
Enter a number: 12  
The number is positive
```



if - else

Q2. Write a program that checks whether a number entered by the user is even or odd.

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

SAMPLE OUTPUT

```
Enter a number: 7
Odd number
```



if - else

Q3. Write a program that finds the greater of two numbers entered by the user.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print(a, "is greater")
else:
    print(b, "is greater")
```

SAMPLE OUTPUT

```
Enter first number: 15
Enter second number: 22
22 is greater
```



Q4. Write a program that assigns a grade to a student based on marks scored.

Marks	Grade
90 and above	A
75-89	B
40-74	C
Below 40	fail

```
marks = int(input("Enter marks: "))
```

```
marks = int(input("Enter marks: "))  
if marks >= 90:  
    print("Grade A")  
elif marks >= 75:  
    print("Grade B")  
elif marks >= 40:  
    print("Grade C")  
else:  
    print("Fail")  
  
if marks >= 90: Grade A  
elif marks >= 75:  
    print("Grade B")  
elif marks >= 40:  
    print("Grade C")  
else:  
    print("Fail")
```

SAMPLE OUTPUT

```
Enter marks: 82  
Grade B
```